PROJECT OVERVIEW STATEMENT	Project Name: Prescription Label Reading	Student Name: Rasika Sanjay Gulhane

Problem/Opportunity:

Support elderly or vulnerable patients should be a focus for many businesses. This is especially true for those in the health care sector. Enabling voice messages can make it easier for elderly people to understand your message. Text-to-Speech can provide peace of mind by empowering you to give better services. For example, you could even send voice messages that read prescription labels.

This can be a real challenge for anyone with reading difficulties, not to mention the elderly and visually impaired. A talking label, sent straight to your device, makes it easy to know everything about your medication. Dosage info can also be tracked and shared with caregivers.

Goal:

The goal of this project is to deploy a user-centric healthcare communication platform with integrated Text-to-Speech technology, enabling the delivery of personalized voice messages and medication information to elderly and vulnerable patients. The **specific objective** is to achieve increase in medication adherence and user satisfaction within six months of the platform's launch. This goal is **measurable** through user feedback, app analytics, and adherence data. The project is **assigned** to the development and implementation, ensuring accountability for its success. **Realistically**, with the available resources, the project will focus on key features for immediate impact while allowing for scalability. The **time-related** aspect involves achieving the stated 20% improvement within the specified six-month timeframe, aligning with the rapid deployment and adoption of the platform.

Objectives:

Objective Statements for the Project:

1. **Outcome: Implementation of OCR Technology**
   Time Frame: Within the first month of the project initiation
   Measure: Achieve an OCR accuracy rate of 95% or above
   Action: Integrate advanced OCR techniques, conduct rigorous testing, and fine-tune algorithms to ensure accurate text extraction from prescription images.
2. **Outcome: Prescription Text Pre-processing**
   Time Frame: Concurrent with OCR implementation
   Measure: Reduction of OCR-related errors by 20%
   Action: Develop pre-processing algorithms to clean and enhance extracted text, addressing OCR errors and improving the accuracy of subsequent processing steps.
3. **Outcome: Medicine Name and Dosage Extraction**
   Time Frame: Following successful OCR and pre-processing implementation.
   Measure: Achieve a recognition accuracy of 90% for medicine names and dosage information
   Action: Employ natural language processing (NLP) techniques and pattern recognition algorithms to accurately identify and extract medicine names and dosage details from the preprocessed text.
4. **Outcome: Integration of Text-to-Speech Approaches**
   Time Frame: Concurrent with OCR and extraction phases
   Measure: Seamless integration of TTS models with minimal latency
   Action: Select and integrate advanced TTS approaches (e.g., WaveNet, Tacotron, or Deep Voice 1) to convert the extracted text into natural and comprehensible speech.

5. **Outcome: User Interface Design**
   Time Frame: Within the first two months of the project
   Measure: Achieve a user satisfaction rating of 80% or above
   Action: Develop an intuitive and accessible user interface that allows users, particularly visually impaired patients, to easily input prescriptions and interact with the application through voice commands.

6. **Outcome: Voice Output Customization**
   Time Frame: Concurrent with user interface design
   Measure: Offer customization options for pitch, speed, and language preferences
   Action: Implement user-configurable settings to allow individuals to tailor the synthesized speech output according to their preferences.

7. **Outcome: User Feedback Mechanism**
   Time Frame: Ongoing throughout the project
   Measure: Collect and analyze user feedback regularly; implement improvements based on feedback
   Action: Integrate a feedback mechanism within the application to gather user input on the accuracy and clarity of the synthesized speech, ensuring continuous improvement based on user experience.

Success Criteria:

1. Accuracy of Medication Information: Achieve a recognition accuracy of at least 90% for extracting medicine names and dosage details from prescriptions within a month.
2. Response Time for Model: Demonstrate model response times within acceptable latency limits, with 95% of requests processed in less than 2 seconds.
3. Robustness of Code and API: Conduct extensive testing to ensure the code is modular, safe, and can handle diverse inputs.
4. Performance Metrics: Meeting defined performance metrics, such as model response time for a given input dataset, with a specific threshold for acceptable latency.
5. Optimized Solution: Demonstrate optimization at the code and architecture levels.
6. Documentation and Collaboration: Successful completion of comprehensive documentation, including a well-maintained GitHub repository with a README file covering project details, execution instructions, and collaborative evidence of effective teamwork.

Assumptions, Risks, Obstacles:

Assumption: Data Quality and Consistency: The quality and consistency of prescription data in the provided Cassandra database are assumed to be sufficient for accurate OCR extraction and subsequent processing. Any discrepancies or inconsistencies in the dataset could lead to challenges in model training and accurate text extraction.

Risk: Integration Challenges: Integration challenges may arise when combining OCR technology, Text-to-Speech models, and database access. Compatibility issues or unforeseen complexities in integrating these components could lead to delays in development and deployment.

Obstacle: User Adoption and Accessibility Testing: The project's success depends on user adoption, especially among visually impaired individuals. Conducting effective accessibility testing and ensuring the system meets the diverse needs of the target audience may pose challenges, potentially affecting user satisfaction and the project's overall success.

Prepared By	**Date**	**Approved By**	**Date**
Rasika Sanjay Gulhane	06 Feb 2024		